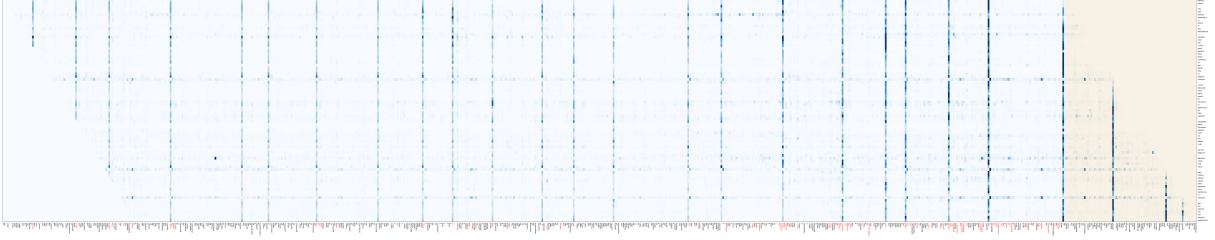
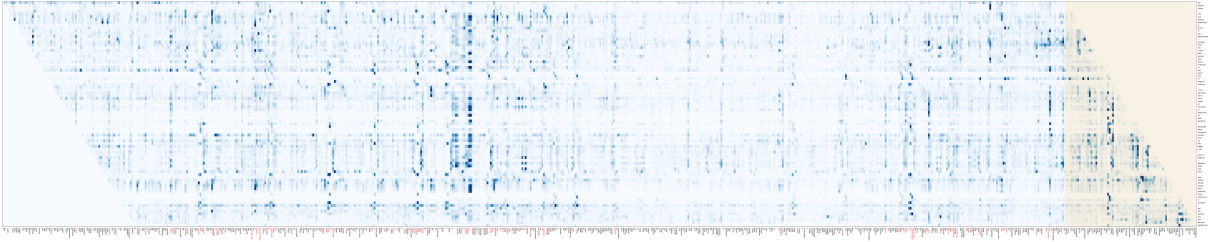


(a) Head 8 from layer 1.



(b) Head 78 from layer 8.



(c) Head 158 from layer 16.

Figure 6: Visualization of the three heads with a wide attention range. Each row corresponds to a target location/token and each column corresponds to a context location/token. Tokens in the memory that have top 20% attention values are highlighted in red.

- For the head 78 in the 8th layer (a middle-level layer), we see a very sparse attention pattern scattered in all ranges of the memory. Again, this well fits our intuition that as information accumulates, the network may focus on some particular position with special interests.
- For the head 158 in the 16th layer (i.e. the last layer), each target location (corresponding to each row) has its own distinct sparse focus, differing from head 78 where target locations largely share the same attentive location in memory. Meanwhile, the pattern is also different from the case of head 8, where a few locations are clearly attended more than others.

Finally, as we have discussed in section 3.3, the attention score can be decomposed into four intuitive terms. Here, we want to further investigate how these four terms contribute to the overall attention trend in Fig. 5. Since the term (c) represents the global content bias, i.e., the prior importance of each word regardless of the context, we will leave it out and focus on the terms (a), (b) and (d). So, for each term, we take the Softmax w.r.t. the memory span and average the resulted distribution of all tokens in the validation set. The results are visualized in Fig. 7:

- Since term (a) is fully content-based addressing, when averaging over all target words, the result is essentially uniform over the entire context, except for a few very close words, which are likely to be semantically similar to the target word.
- The overall trend of term (b) highly resembles that of the entire attention distribution in Fig. 5. It suggests that the global trend of focusing on the nearby context is largely contributed by this content-dependent positional bias.
- The overall trend of term (d) is also focusing more on nearby words. However, compared to the trend of term (b), it is clearly flatter and biases towards a longer context.